Fighting Food Waste with Artficial Intelligence

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Project for Introduction to Artificial Intelligence

Abstract

While supply chains of hospitality businesses often are still managed manually, human error regularly results in food being wasted. Could this be avoided using Artificial Intelligence? This report focuses on posing a possible solution, which consists of an intelligent program that could learn to manage the supply chain of hospitality businesses more efficiently than humans in order to reduce the waste of food, money and resources.

Keywords— Food Waste Reduction, Artificial Intelligence, Machine Learning

1 Introduction

Each year, an estimated 51 million kilograms of food is wasted by dutch hospitality businesses alone. Throughout the production, industrial processing, transportation and storage of food, carbon and methane gasses are emitted, which greatly contribute to climate change. If this food then ends up being unused, the waste disposal process accounts for additionally emitted greenhouse gasses. As there also is an unethical aspect about the wasteful treatment of food while hunger and starvation still exist on earth, it also is not in the financial interest of hospitality businesses. For our project in the realm of Artificial Intelligence, we want to focus on an idea for reducing food waste in hospitality businesses. We will elaborate on the following research questions.

- How would a supply chain prediction system work?
- How much and which data is needed to make an efficient predictive system?
- What algorithms would be useful in a predictive system?
- Would a predictive system be able to outperform human supply chain managers?

With this report we aim to make a contribution to the global effort of reducing our greenhouse gas emissions and food waste. Therefore, pairing environmentally-friendly business practices with a financial incentive could be effective in making hospitality businesses participate in protecting the climate and avoiding wastefulness.

2 Domain and Task

2.1 Domain

This project will predominantly focus on the domain of food waste. This is due to the fact that some of our group members work in the hospitality industry. It has been established that there is a need for the optimization of the food ordering process in a socially conscious way. The food catering service is responsible for 14 percent of all wasted food. [8].

2.2 Task

In order to prevent the catering industry from wasting food due to ordering surpluses, we have come up with the idea to use an AI-system to predict the amount of product needed for each given period of service.

2.3 Problem

The several issues within the domain of food waste in the catering industry that were found from research were the following:

- Food supplier ordering surpluses.
- Lack of distribution infrastructure for expired food.
- Government laws that causes non-expired food to be wasted.
- Inadequate food storing practices.
- Overproduction of food orders.

In this paper the problem domain that will be focused on is in the realm of food ordering surpluses. It often occurs that either too much or too little food has been ordered by food-related businesses. This will either lead to more wasted food, or dissatisfied customers who are not be able to order their preferred dish. In both cases, the business is losing funds and could be working more efficiently. On top of that, food waste has a strong negative impact on the environment. [12]

2.4 The input and output description

The input into our AI-system would include the ordering data of restaurants from the past years and basic data about the eating out habits of humans, figuring in metrics like holidays, birthdays and other activities that commonly involve eating at restaurants. The output of our system would be predictions about how much food will be needed in each week for individual restaurants, helping them to save money, save valuable food resources and help to eliminate food waste. Nevertheless, if a restaurant has a surplus of food at the end of a day or week, our AI-system could help food banks to get into contact with them to rescue food from being thrown away.

2.5 The knowledge that is needed

For further development and implementation of this idea, we will need more information from the restaurants to which this AI would apply. Firstly, we need to know what percentage of food-oriented hospitality businesses would want to apply AI to their finances and work towards preventing food waste. We should gain information on how restaurants track their income and expenses around their orders and how much financial incentive they would have to use an AI to decide their orders. We have already contacted people we know that work in the hospitality industry, so they could give us more insight about whether our application would be useful.

3 Literature and knowledge acquisition

3.1 Literature

Ten papers were collected that could help to build the best solution to the domain of avoiding food waste at hospitality businesses. Most of them were found using Google Scholar, searching the following terms:

3.1.1 Terms:

- Food waste hospitality
- Food waste hospitality artificial intelligence
- Food waste hospitality inventory OR management
- Food waste restaurants
- Food waste reduction
- Food waste reduction AND artificial intelligence
- Food waste reduction AND machine learning
- Food waste AND global warming

3.2 Presentation of suitable literature

Below is displayed the title, link, a short summary of the content and a conclusion about why the information displayed in the texts could benefit us.

3.2.1 Institutional sustainable purchasing priorities: Stakeholder perceptions vs environmental reality [9]

Summary:

The research investigates how big business deal with cost effectiveness of food orders and also the sustainability of this practice.

Conclusion:

It is relevant to our project to understand how sustainable food orders are from suppliers to catering services from a stake holder perspective.

3.2.2 "Back of house" – focused study on food waste in fine dining: the case of Delish restaurants [1]

Summary:

This research looks at compares how food supplier shortages impact overall food waste in restaurants.

Conclusion:

The contracts that restaurants have with food suppliers prevents them from being able to make efficient food order quantities.

3.2.3 A Suitable Artificial Intelligence Model for Inventory Level Optimization [2]

Summary:

This paper discusses the possibilities concerning AI in the supply chain management. It states the specific types of AI needed to create an accurate prediction, and what types of data it would need.

Conclusion:

This could assist us in gaining a much better and detailed understanding on the specific elements which would be needed to create the predictive AI described earlier.

3.2.4 Minimizing Food Waste at Google: Creating Production Innovation and Purchasing Practices [11]

Summary:

This research explains the attempt of a large tech company (Google) to reduce the environmental impact of its food waste.

Conclusion:

The information gained during this process could be very much of use during our project. To see what has previously worked or what obstacles Google encounter during this time could be crucial for the decisions we make for our project.

3.2.5 IoT edge computing in quick service restaurants [10]

Summary:

This paper investigates the fast-food restaurants and gives them important and helpful information about the data patterns and thus tries to minimize the food waste.

Conclusion:

This could really help us as it also gives information about fast-food restaurants and kind of has the same idea as we have. So, we might learn from this as well.

3.2.6 A survey of machine learning techniques for food sales prediction [3]

Summary:

This paper is about what machine learning approaches have already been used for food sales prediction. It also discusses the difficulties and possibilities for applying machine learning to food sales prediction.

Conclusion:

This paper will give us intel on what has already been looked into and things that just are not possible to do for us. It might give us some ideas as well, which is good for the creativity.

3.2.7 Artificial intelligence in the design of the transitions to sustainable food systems [4]

Summary:

This is a literature review about how AI could impact the transition to a more sustainable food system. It focuses on a few approaches by the authors of the reviewed books.

Conclusion:

It could help us navigate through different ideas and approaches as how to develop an AI algorithm that will actually benefit the businesses using our services.

3.2.8 Food waste management innovations in the foodservice industry [7]

Summary:

This paper focuses on reducing food waste by implementing new radical innovation in the foodservice sector. It offers several (theoretical) approaches for how to tackle various wastage problems in this industry.

Conclusion:

We could use this paper to gain a more theoretical and scientifical approach on our domain.

3.2.9 Food waste management in hospitality [6]

Summary:

The paper looks at how food waste in the hospitality industry can affect the society at large. Also, how the in-house training and investment costs could contribute to savings for a hospitality business.

Conclusion:

This paper can be used to understand how food waste impacts the socio-cultural standpoint of the environment.

3.2.10 Food waste in the hospitality industry having a global environmental impact [5]

Summary:

This paper looks at how the cultural production of food in Malaysia impacts how it interacts with the country's hospitality industry and how that can impact the ecosystem in the country as a whole.

Conclusion:

This paper can be used to look at how food waste in hospitality is impacted outside of western European countries and how the application we make could be used on a global scale.

3.3 Knowledge acquisition

3.3.1 What experts have we spoken to?

We decided it would be very useful to interview restaurant owners / experts in the field of running a culinary business. These people would have knowledge and experience with regards to inventory management, product estimation and employee time management. They could supply us with information on the need of our AI and potential problems we might have overlooked.

We were able to arrange an interview with restaurant owner Ron Blaauw. Ron owns multiple restaurants in Amsterdam and Ouderkerk aan de Amstel, one of which has a Michelin star.

We were also able to arrange an intervieuw with Björn Vlegels, Chef at the Blauwe Draak in Amsterdam. However, due to the current coronavirus pandemic, he was not available for an in person interview, so we conducted our interview via email.

3.3.2 Which aspect of knowledge elicitation did we use?

We conducted two interviews with experts on the area of restaurant management. This means we have used an individual focused method. Below, you can see the questions we asked during both interviews. During the interview with Ron Blaauw we were also able to ask follow up questions on his responses.

3.3.3 The questions

1. How often do you order fresh products?
2. How do you decide how much to order of each product?
3. Do you keep track of how much of each product you have left (at the end of the day)?
    - How often?
    - How?
        - How long have you been using this method?
4. Do you ever have to throw away food that has gone bad?
    - How?
    - Do you try to minimize this?
        - How?
5. Do you think legal regulations on waste management and hygiene for businesses contribute to the problem?
6. Do you think a computer program could decide how much to order?
7. If a computer program were to attempt this, what would be the most important factors to keep track of?
8. If this would work and minimize food waste and costs, would you use it if other restaurants can use it as well?
9. Are you willing to keep track of all the data if this would minimize food waste and costs?

3.3.4 Interview results

Below we will present the most relevant, surprising and informative information which we were able to gather from the interviews.

Ron Blaauw

The restaurants owned by Ron Blaauw operate very differently compared to the much more common middle class restaurants. His restaurants are much more focused on quality above quantity, and work with a smaller amount of albeit fresher and more expensive ingredients. Knowing this, we still hoped to gain insights into how restaurants operate and how they manage food waste.

Each restaurant (chain) tries to limit food waste in their own way. In the case of Ron's restaurants, they use an inventory management system where they use two refrigerators and are constantly moving products from the one to the other. They also have the chef recommend certain dishes to customers as they're deciding what to order, to convince them to order a dish with ingredients which would otherwise expire. Each restaurant has different resources and possibilities to manage inventory, forcing each different establishment to "reinvent the wheel" on the area of food waste management.

Ron predicted that 3 out of the 50 people working for his restaurants could be replaced by an AI which manages inventory.

Björn Vlegels

One of the ways Björn Vlegels manages to decrease food waste is by giving food which is about to go bad to his employees. This way a large part of the food which would have gone bad, is still able to be eaten. However, this is still not financially optimal for the restaurant. His personnel also spend a lot of time tracking the food in inventory by hand. This system has been noted to be venerable to human error, and will as result of that regularly have food waste due to miscounting or misplacing.

4 Computational solution

4.1 What data is being collected?

The AI needs to collect two main types of data to make an accurate prediction.

4.1.1 Establishment specific information

The establishment specific information (ESI) contains the information which can only be collected by the establishment itself. This contains the following:

- How often and on which dates the establishment wants to order products from its suppliers(s)
- The amount which has been ordered per product (either in kg or in number of products)
- Regular updates on the number of products still in stock
- The ingredients and amounts of products used to produce each order (standard quantities, e.g. Burger: 100g meat, 1 white bun, 1/6 of a tomato, etc.)
- The number of placed and completed orders, containing information on time, date and variations (e.g. extra tomato) for each order

A large part of this information, if not all, is already being kept track of by many establishments for the sake of their own bookkeeping. The flowchart describing the general algorithm is displayed in Figure 1.

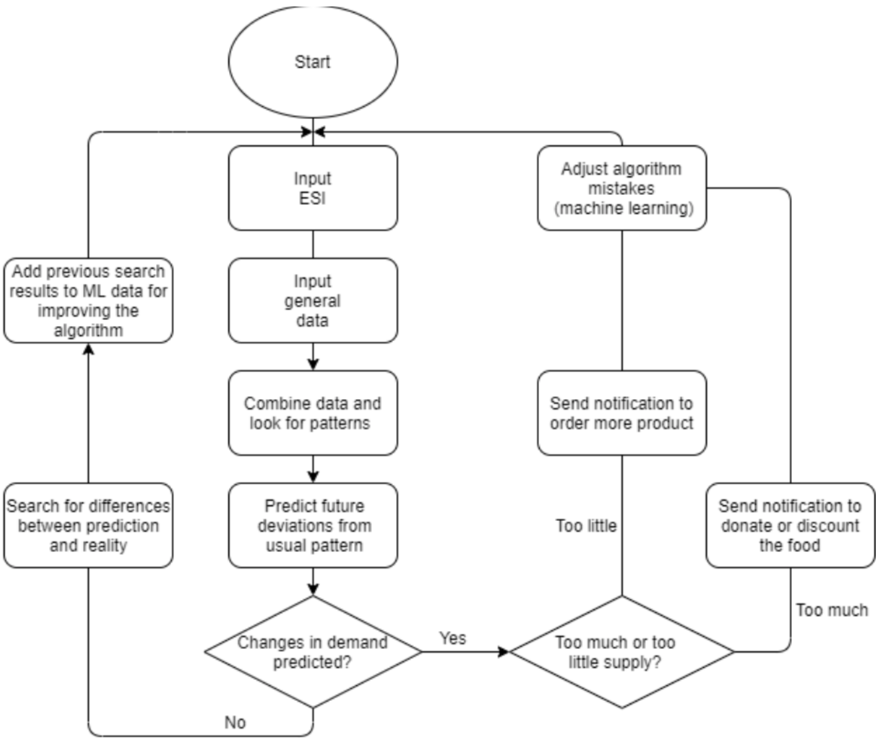

Figure 1: Detailed parts of the system

4.1.2 General information

The number of customers visiting an establishment at any given time is partially predictable with ESI described above. However, there are also many external factors which can lead to a variation in customers. Depending on the accuracy of the AI and the accessibility of information, more external factors can be taken into account at a later time. At the moment, the most important information to keep track of is:

- Keeping track of the date is vital for this AI. Simply making the connection that there is more demand on a Saturday than on a Wednesday, allows the AI to create a better and more accurate representation of the reality.
- A list of dates (or date ranges) with spikes in demand. On some days there's a very significant spike of people visiting catering businesses, e.g. Valentine's day or the week where summer vacation starts, etc.
- Gradual yearly trends. These are changes in demand which are slower and less drastic than the dates with spikes, but last longer. This would contain things such as vacations (considering tourists) and birthday trends (more birthdays in the summer than in the winter).
- The current and forecasted weather. Weather plays an important role for people when deciding whether or not to eat out. This is especially the case for establishments with outdoor seating.

4.2 How is the data being collected?

The ESI can only be collected by the establishment itself and its employees. Logging incoming and outgoing inventory is already something almost all establishments must do, and therefore already have employees trained and scheduled to do so. However, this is not always digital. For the AI to be able to use this information, it needs to be entered into a digital user interface, created to allow the AI to store this information in the most efficient way. Information on average ingredients used per order only has to be entered once by hand. Keeping track of the amount of orders and variations is something which is already done by almost all establishments, for keeping track of the bill. Simply creating an overlay or some way to connect this system to the AI, will allow the AI to collect this information while the employees would not need any training or more time.

The general information is publicly available and can be collected by the people programming the AI. By adding a weather forecast API, it will allow the AI to read up to date weather forecasts for a coming time period.

4.3 Which algorithms are being used to analyse the data?

One of the main algorithms that is being used is a supervised learning algorithm that creates a snapshot time period. Within this time period there are lagged number of variables that determine the size of the snapshot window. Time granularity is also used to determine default values. An example of these default values would be the following:

- Monthly sales data
- Weekly sales data
- Hourly sales data
- Day of the week
- Time of day
- Which quarter in the year?
- Which season in the year?

All of the above variables would serve as probable default values for training data within the algorithm. Temporal granularity is when daily sales predictions are used for food produce with a short shelf life. For example, milk, cheese, yoghurt etc. For products with a long shelf life, weekly predictions are made which will help with optimisation for stock management. Then, further predictions will be made to help the management team optimise for monthly and quarterly assessment of future stock orders.

Within the subset of propositional learning the Support Vector Machine algorithm (Figure 2) can be used to separate the supermarket data into different classes. By creating an estimation line between data points, the algorithm takes in the input data and creates the separation line with the output data. Along with this, an auto-regressive moving average(a model that shows the difference between the errors and prior observations) can be used to model the behaviour in certain produce sales around different times of the year. This model takes the past sales history and forms data points from it, which are called "lagged observations". Then, this is used to forecast future observations in a time series. Lastly, weights are created and applied based on how recent a sale was. The auto-regressive process regresses a variable on past values of itself. For instance, past sales data for cheese forming a model for future predictions of cheese sales, from whole sale orders to a restaurant. When a non-deterministic algorithm or other algorithmic behaviours are removed, error terms of previous time points can made into a function and used for calculation of the current observed values.

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4.3.1 Comparisons to other algorithms

There are three main methods of a machine learning algorithm. For this project, supervised learning is the most suitable. When a supervised learning algorithm processes input data into output data, it will learn from the difference between the generated output data and the correct output data. If those two differ too much, then the AI will change the way that it processes the data, with the hope of achieving a result closer to the correct output data. This would mean that for each time period, the AI would predict the amount of food which needs to be ordered and will at the end of the time period, compare that predicted value to how much food was actually needed. The AI can train this on old data and

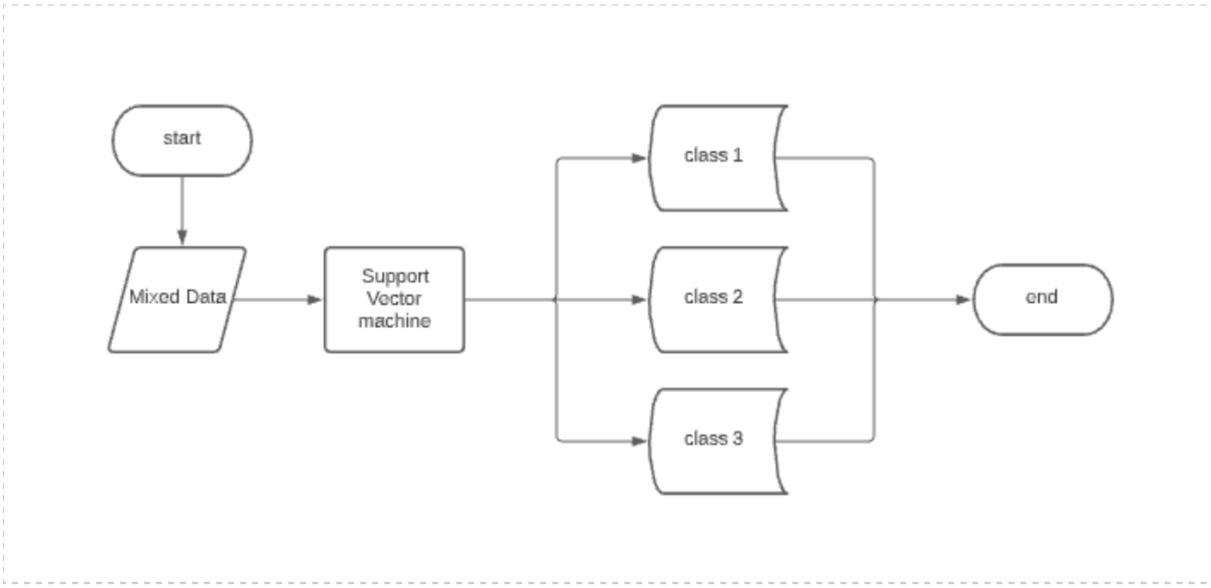

Figure 2: Support Vector Machine

can also improve itself continuously when being used by an establishment. [13] An alternative learning method would be unsupervised learning. This would not be a useful learning method for this situation, as it is meant to find patterns, without ever learning or being guided to produce a certain desired outcome. This means unsupervised learning might find an interesting link between 2 factors (weather and meat consumption), but it will not produce nor train to produce an accurate prediction of product demand.

The final alternative would be reinforcement learning. An example of what reinforcement learning is capable of would be video game AI. If a computer-controlled character keeps losing in a virtual fight, it will keep trying new inputs and combinations until it wins. Even though this learning method does learn and adjust itself to past events, it is based on attempting random actions to improve it's "score". This would not be suitable for the AI we have designed as it should not be based off of random actions.

4.4 Which results are being computed?

The results that are being computed are the buying habits and increase in sales of users over a specific amount of time. This data is then used to predict future orders and to optimise the spending on wholesale orders from suppliers.

4.5 How are the results being computed?

For every consecutive order, the customer will need to input in the ingredients just once and then the AI will be able to determine what that order is made of. Once all of the orders have been filled in with their ingredients, the AI will calculate what the restaurant needs to have in stock in order to make dishes. As you can see in the previous paragraph, the buying habits of the customer need to be known like; which ingredients and which supplies the restaurant needs. It will also take into account an increase in sales over time, therefore we have to get data at different sales spikes. For example, if the restaurant sells more than they usually do in busy times like; the summer, bank holidays, religious holidays like Christmas or Easter. As the AI would have learnt about the yearly agenda, it will also know when certain holidays happen and take that into account in the prediction to makes sure it orders more than the usual.

4.6 How will this be presented to the user?

In order for the target group to benefit from our AI-system, it would need to fit seamlessly into their day-to-day processes. This means that it must not pose an obstacle to the established daily routines nor, cost too much time to use. To make our system provide the best possible predictions for our users, a lot of ESI is required. The needed information is extracted from accounting and bookkeeping data of the previous years. This means that our system works best with businesses that have had digital tracking systems in place for years. Businesses that either haven't used this kind of system in the past or recently founded businesses may not be well suited for the usage of our systems, as there is time cost in paper to digital data conversion or not enough data to make helpful predictions. Considering that most of our users will have a digital tracking system for that reason, it could easily be connected to our AI-system. The design of our AI-system should be similar to the most common digital tracking systems, as the employees of the businesses are already familiar with it.

Generally, the predictions the system makes should be presented in a calendar-like format, displaying the probable supply and demand, as well as suggestions for ordering new product. Also, there should be push-notifications for when the business is close to running out of a certain product, so it can be reordered in time.

5 Experimental evaluation

5.1 Application description and key evaluation points

In order to evaluate our strategy for eliminating food waste in the hospitality sector, we must first develop a Conceptual Model and Key Evaluation Points. They are used to explain to the user the inner workings of our algorithm, so that they are more qualified to answer our question and evaluate the system.

5.1.1 Application description

Our program first receives input about general data of recurring and predictable events that lead to more restaurant visits. Then the ESI (establishment specific information) is added. It consists of supply and demand data of the previous years. Then, the data is combined and scanned for patterns. Afterwards, the algorithm will be able to predict future deviations from the usual patterns, considering the general data about fluctuations in restaurant visits.

If the algorithm detects a future deviation that the supply chain is not able to compensate naturally, a notification gets sent out to the user to inform them to alter the supply chain (order more or less product).

Over time the algorithm is supposed to get better at predicting changes in demand.

5.1.2 Internal evaluation

For the internal evaluation we will use logged inputs from the past (e.g. the sales from the month of April 2019) and having the AI predict how much to order for another date in the past (e.g. the first week of May 2019). This way we can evaluate the AI "in the lab" before implementing it into the real world.

- Effectiveness
  - We will compare the human predictions for that week to the predictions made by the AI and which one is turned out to be closer to the real measured value of what was actually needed.
- Usability
  - We can test the functionality of the user interface with a wide range of people to test if it is intuitive and requires no extra training or time investment.
- Actual benefit (monetary, environmentally)
  - Based on the result of effectiveness, we can make estimations of the monetary and environmental effect of the AI by summing the value and carbon footprint of the food which would have been wasted without the AI.
- Prediction accuracy (changes in supply and demand)
  - Instead of comparing the predicted value to what was predicted by the humans, we will compare the predicted values to what was actually ordered.
- Consistency
  - We will check if all the values above stay consistent over time and don't fluctuate dramatically.

5.1.3 External evaluation

- Effectiveness
  - By measuring the average food waste over a certain time before implementing the AI and comparing it to the average food waste over the same time period, we can make a general statement on the effectiveness. This will essentially be the difference in how close humans were to the actual used amount of food compared to how close the AI is.
- Usability
  - We can ask the employees of the establishment if they experienced any issues in usability and perhaps ask the manager if there was any difference in time effectiveness of the employees.
- Actual benefit (monetary, environmentally)
  - This will be tracked by forming a comparison between the customers previous waste/spending ratio and the present ratio in order to calculate exactly how much money is being saved.
  - Ask the restaurant users if they were able to eliminate waste.
- Prediction accuracy (changes in supply and demand)
  - Instead of comparing the predicted value to what was predicted by the humans, we will compare the predicted values to what was actually ordered.
- Consistency
  - We will check if all the values above stay consistent over time and don't fluctuate dramatically.

5.2 Evaluation questions and measurable outcomes

As we are yet to develop the application, we can only ask theoretical stakeholders about their opinion concerning our program. In order for them to give valuable feedback, we have to make them familiar with the application. This can be done by explaining the algorithm, giving an example of the data processing and showing a mock up of the application. Then, the target group can answer following questions:

- Do you think this program could benefit you financially?
- Do you think you would use our program?
- Do you think the program could effectively help to eliminate food waste?
- Do you think you can make accurate predictions about supply and demand by yourself?
- What possible obstacles could hinder you from using our application?

5.2.1 Measurable outcomes

For measurable outcomes, we need hard data. This could be the simulated data from our application model or real data from customers. Measurable outcomes in our case would be:

- Prediction accuracy:
  - Comparing the predicted versus the actual supply/demand for a product on a specific day or week to see whether the algorithm was helpful.
- Monetary benefit:
  - Is less money being spent for the same amount of served food and drinks than before the algorithm was in use?
- Error margin:
  - How often does the algorithm make a specific mistake before detecting and avoiding it?
- Amount of data:
  - How much user-input data is needed to make the algorithm work best?
  - How much general data is needed to make the algorithm work best?

5.2.2 Evaluation design

Proof of concept The subjects who will be studied are members of our target group. Our target group consists of the owners of restaurants, supply chain managers at hospitality businesses and other staff members. All types of stakeholders of our application should be asked for their opinion to get an all-encompassing feedback. We are going to speak with the interviewees this week, who will provide us with insight on the effectiveness and usability of our application. Our methodological approach will be an inductive research approach. Therefore, we will ask our target group questions to identify strengths and weaknesses of our program. The interviews will be held in Amsterdam.

Testing the algorithm To test the algorithm, we can either work with historic hard data from a restaurant or use the fictional data from our application model. We can input half the data into the algorithm and let it predict what the future supply and demand would look like. Then, the rest of the data would be used to compare the future predictions of the algorithm to the historical hard data of the restaurants. The algorithm should be trained this way and could also be evaluated this way, as the prediction accuracy is easily derived from the results of such tests. An indicator if the algorithms usability is, whether the algorithm can outperform supply and demand predictions by a human Supply Chain Manager. If yes, the algorithm can be applied in real businesses to test further. The restaurants should be able to detect the benefits of using the algorithm clearly.

5.2.3 Data collection

The data we collect will be from our interviews. It will consist of subjective opinions about our application. Also, we can either use fictional, historical or even real data from restaurants to test the algorithm.

5.2.4 Data analysis and presentation

After we conducted the interviews, we will prepare the answers we got, to get an overview. We will present the common answers we got to each question. The results of testing the algorithm will be reviewed, displayed and explained.

6 The Conclusion

Our project focused on building an AI-system that could help prevent food waste in the hospitality industry. At first, we thought about what our domain was and on what part of the problem we wanted to focus on. Then we started doing research on the domain to find out ways that an AI-system could help to prevent unnecessary food waste. For this, we also interviewed stakeholders of our idea. In order for us to establish a computational solution, we had to first specify the kinds of data that we would need. With this information, we tried to find different algorithms that would enable us to incorporate all vital elements important for our system. From this we learned how different mathematical concepts can contribute to different supervised learning algorithms. In addition to this, we learned the difference between different algorithms in machine learning overall. Especially the way adjustments are made to the algorithm after detecting a previous mistake, is crucial to the functionality of the system. After we had designed a concept for the system that fulfilled all of our requirements, we then needed to evaluate its efficiency. This has to be done through internal and external evaluation. We had no way of completing the external evaluation, as we have no existing version of our system. Therefore, we could not form a valid conclusion, about whether or not our system would be applicable for daily business practises.

Though we were not able to conclude our evaluation due to the lack of the complete system, we therefore can only make assumptions about the quality of our idea. We think that there is a chance that this system could be useful in preventing food waste. We evaluated it to the best of our abilities, but of course we can not proclaim that we did not miss some of our own errors. During the execution of this project, we gained much more knowledge in the domain of food waste, and have learnt much about different AI systems and methods. For each group member, this was also their first experience with writing in Latex.

We hope to next time be able to dive deeper into the programming part of the AI, as all of the group members are interested in the technical aspects of making a working system. Furthermore, we would like to conduct a more comprehensive research within the catering industry. This could be in the form of interviews or surveys. With this we hope to gather a better understanding of the need for an AI system, and the challenges we would encounter. Finally, we would like to expand our research on AI systems for a more complete understanding of the methods we could apply to solve our problem.

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